$$\phi^{\in \Phi} ::= \mathrm{FCall}(f) \mid \mathrm{Const}(c) \mid \mathrm{Glob}(x) \mid \mathrm{Param}(x) \mid \mathrm{PConst}(c)$$

0

---

transfer functions

$$T :: \mathrm{Stm} \to \hat{\mathrm{Sto}} \to \hat{\mathrm{Sto}}$$

$$\hat{\rho} \in \hat{\mathrm{Sto}} :: \mathrm{Var} \to 2^{\Phi}$$

$$\rho :: [x_1 \to \{\}, x_2 \mapsto \ldots$$

$$x_3 \to \{\mathrm{Glob}(y_1), \mathrm{Param}(y_2)\}]$$

$$\hat{h} : \hat{\mathrm{Heap}} :: _____ \to _____$$

?

Assignment - Constant

$$T(x = c, \hat{\rho}) = \hat{\rho}[x \mapsto \{Const(c)\}]$$

Assignment - Constant (let)

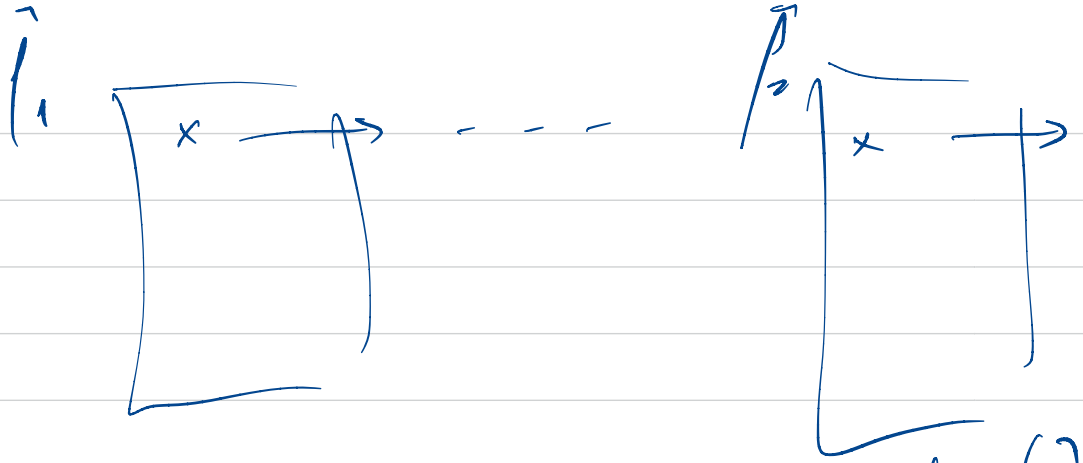
$$T(\text{let } x = c, \hat{\rho}) = \hat{\rho}[x \mapsto \{Const(c)\}]$$

Assignment - Fun Call

$$\Phi_{args} = \bigcup_{y \in \bigcup_{i=1}^{n} vars(e_i)} \hat{\rho}(y)$$

$$T(\text{let } x = f(e_1, \dots, e_n), \hat{\rho}) = \hat{\rho}[x \mapsto \{Fcall(f_1)\} \cup \Phi_{args}]$$

$$\begin{tikzcd}
 & 0 \arrow[dl] \arrow[dr] & \\
\hat{f}_1 \quad 0 \arrow[dr] & & 0 \quad \hat{f}_2 \arrow[dl] \\
 & 0 &
\end{tikzcd}$$



$$x \in \underbrace{\mathrm{dom}(\hat{f}_1) \cap \mathrm{dom}(\hat{f}_2)}$$

$$(\hat{f}_1 \sqcup \hat{f}_2)(x) = \left\{ \hat{f}_1(x) \cup \hat{f}_2(x) \right.$$

$f(\widehat{db}, x)\{$ $\longrightarrow \hat{\rho} = [\,]$

$\quad if(*)\{$

$\quad\quad db.banana = 2$ $\longrightarrow \hat{\rho} = [\,]$

$\quad \}$

$\quad \vec{\ } let\ y = \widehat{db.banana}\ *$ $\longrightarrow \hat{\rho} = [y \mapsto \{ \underline{Panam(\overset{id}{db})} \}]$

$Panam(db)\ \boxed{eval(\text{"}x = \text{"} + y)}$

$\}$ $\longrightarrow$

function ( ) {

id: let x = readSthFromQueryString();

eval(" banana = " + x)

}

FCall(RSFQS)

id

} = [FCall(RSFQS)]

```
function (db) {
  db.foo = 3
  eval("banana =" + db.foo)
}
const(3)
```

$\Rightarrow$ ✗